



Module 11 Day 2: Cryptography

Activity File: Optimizing with Asymmetric Public Keys

In this activity, you will continue to play the role of a security analyst for the Hill Valley Police Department.

- Hill Valley's largest police divisions, the SWAT team, the Canine Unit, and Internal Affairs, are currently using symmetric keys to exchange secure messages with each other.
- As a security professional, you recommend Captain Strickland seriously consider moving to asymmetric keys, as it would mean much fewer keys to manage.
- Captain Strickland is excited about your recommendation, but would like to understand exactly how many fewer keys each division would be managing with this change.
- You must provide a summary report showing how many keys are saved by moving from symmetric key cryptography to asymmetric key cryptography for the three divisions.

Instructions

1. For each of the following Hill Valley Police divisions, calculate how many asymmetric and symmetric keys will be required for secure communication based on the number of officers.
 - SWAT team: 10 officers
 - Canine Unit: 25 officers
 - Internal Affairs: 45 officers
2. Document how many fewer keys will be required after moving from symmetric to asymmetric cryptography for each division.
 - For example: *The SWAT team will need X fewer keys after moving from symmetric to asymmetric cryptography.*